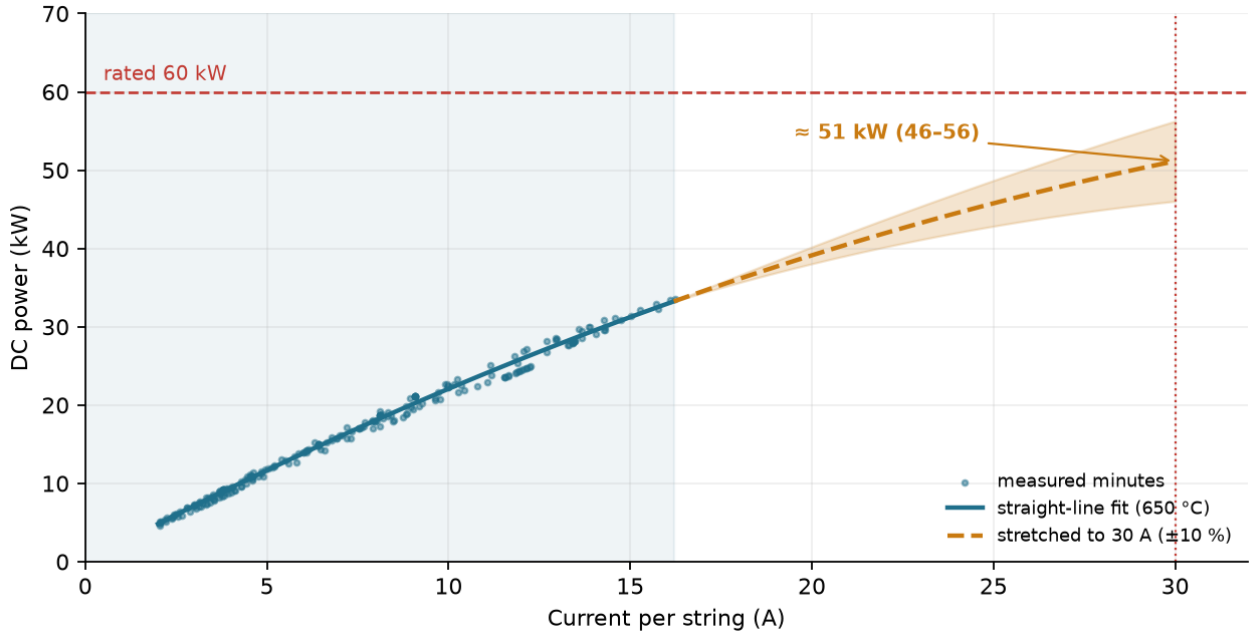
Rated power extrapolation — SOFC plant

Can the part-load data tell us what happens at 60 kW? · 2 November 2026

1. Can the data reach 60 kW?

> **Partly.** Stretching the measured data to the rated current of 30 A gives about **51 kW**, not 60 kW. The voltage and the fuel numbers can be stretched with reasonable confidence. The heat, the exhaust and the safety trips cannot.

The plant never got past half load. The best moment was 33.6 kW at 16.2 A per string on 25 March 2026, 56 % of the rating. Every run was short (8–44 minutes) and most ended in a safety trip. Anything said about 60 kW is a stretch of a line beyond the measurements (extrapolation).



Dots: every load minute in the 39 logged days. Solid: straight-line fit at 650 °C. Dashed: stretched to 30 A with a ±10 % band.

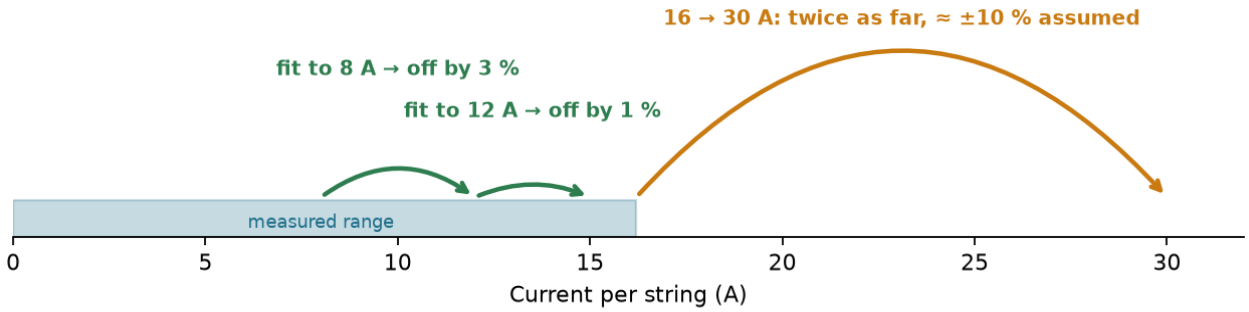
Quantity	Measured best	Rated	Stretched to 30 A
Current per string	16.2 A	30 A	30 A
Voltage per string	≈ 416 V	400 V needed	≈ 341–350 V
DC power (5 strings)	33.6 kW	60 kW	≈ 51 kW (46–56)

Why short of 60 kW: each cell's voltage drops as more current is drawn. The measured drop is about 5 V per amp per string. At 30 A that leaves about 345 V per string, while 60 kW needs 400 V.

If you remember one thing: the data point to about 51 kW at full current — a stretched estimate from half load, good for voltage and fuel, not for heat or safety trips.

2. What can be stretched, what cannot

Numbers that follow a physical law (voltage, fuel) stretch well. Numbers set by the operator or by safety settings (cooling air, exhaust trips) do not.



Back-test: a line fitted on the lower currents predicted the higher measured currents within 1–3 %. The jump to 30 A is longer than anything we could test.

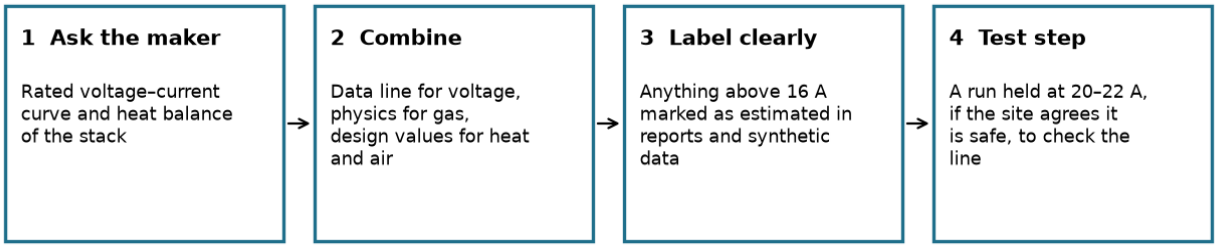
Quantity	Can we stretch it?	What the data say
Voltage and power	Yes, with a band	≈ 51 kW at 30 A (range 46–56). Back-test errors were 1–3 %. Measured voltage follows a straight line; no bending yet.
Natural gas needed	Yes, by physics	Fuel use follows a fixed law (each gas molecule gives a fixed charge). The data match it: 43 % of the gas used at 16 A with 160 ln/min. At 30 A the same gas would be 81 % used, so ≈ 160–175 ln/min is needed for a normal 75–80 %.
Weakest cell voltage	Yes, and it says stop	The weakest cell would fall to ≈ 0.59 V, below every protection limit used (0.72–0.90 V). With today's 0.72 V limit the current stops near 15 A, as seen.
Stack heating	Only roughly	Heat made inside the stack grows about 2.5 times from 16 A to 30 A. The cooling air stayed at 9 000 ln/min in every run, so how the plant cools itself at full power was never seen.
Exhaust and safety trips	No	The exhaust hit its 200 °C trip at half load. The limit was raised to 300 °C on 27 March, after most runs. Trips depend on settings, not on a trend.
Steady full-power running	No	The longest run was 44 minutes and the stack was still warming. There is no settled state to stretch.

ln/min = normal litres per minute. The weakest-cell 0.59 V is a warning sign, not a precise value (the controller holds that cell at its limit while the stack warms).

If you remember one thing: trust the stretched voltage and fuel numbers within about ±10 %. Do not use the data to predict heat, exhaust or trips at 60 kW.

3. What to do next

Use the stretched line as a first estimate of about 51 kW, and anchor it to the stack maker's design data before anyone relies on it.



• **The maker's curve is the best anchor.** Our line covers 2–16 A; the maker's curve shows whether the voltage bends down near 30 A.

• **Mixing sources is normal.** Voltage from our data, gas from physics, heat and cooling air from design values.

• **Labels stop estimates turning into facts.** The synthetic-data generator should leave out anything above 16 A, or tag it as estimated (open decision D3 in the part-load notebook).

• **A test step is cheap evidence.** Holding 20–22 A for an hour adds the missing settled state and checks the line. It needs the weakest-cell limit and the 300 °C exhaust limit agreed with the site first.

Source: data_analysis/part_load_analysis.ipynb (sections 4.1, 6, 7.2) and data_analysis/part_load/. Back-test and chart points recomputed from the same 234 load minutes.

If you remember one thing: the data give a first estimate, the maker's design data confirm it, and only a real test at higher load can shrink the uncertainty.